Diabetic ketoacidosis

Diabetic ketoacidosis

- Serious complication of diabetes mellitus due to an absolute or relative deficiency of insulin and an increase in the levels of insulin counterregulatory hormones (glucagon)

- This is most common in individuals with type 1 diabetes but can occur in those with type 2 diabetes as well. Younger age is also a risk factor.

- The most common precipitating factor for DKA is intercurrent illness, such as infection, trauma, surgery, or myocardial infarction.

- Excess production of keto acids: (acetoacetic acid, hydroxybutyric acid and acetone)

Ketone bodies, an alternative source of energy

- Physiology:
  - They are produced from acetyl coenzyme A mainly in the mitochondria within hepatocytes and used peripherally as an energy source when glucose is not readily available. (e.g. The brain cannot utilize fatty acids for energy when blood glucose levels become compromised.)
  - They are always present in the blood and their levels increase during fasting and prolonged exercise.

- Diabetic ketoacidosis:
  - A high concentration and it is a cause of metabolic acidosis with high AG.
  - Fatty acids are degraded by β-oxidation faster than the formed acetyl coenzyme A can enter the Krebs cycle. Excess acetyl coenzyme A gives ketone bodies

Insulin and glucagon

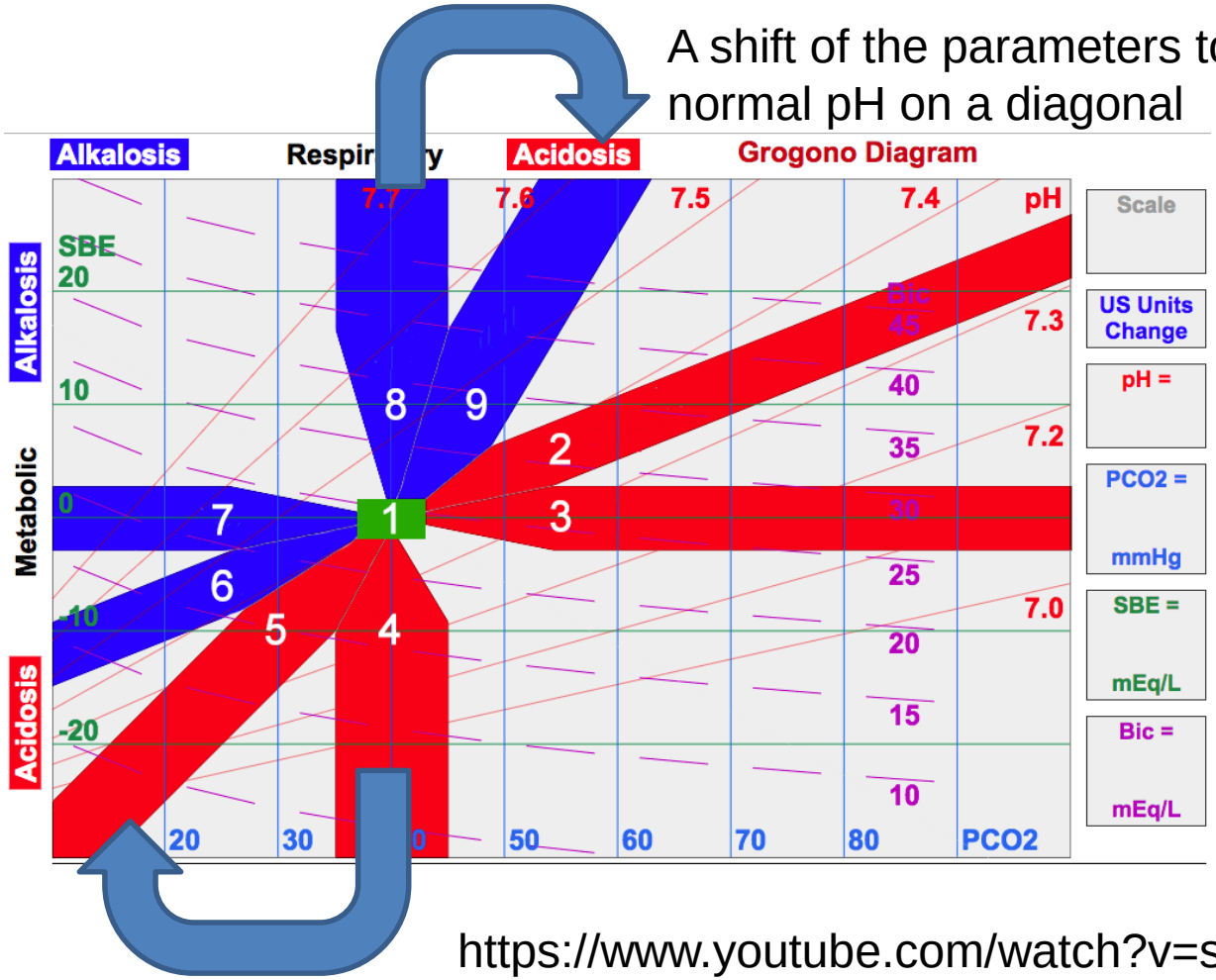
- Physiology:
  - Insulin
    - stimulates glycogenesis and glycolysis
    - inhibits lipolysis, gluconeogenesis and glycogenolysis,
  - Glucagon
    - increases blood glucose level by stimulating glycogenolysis and gluconeogenesis in the liver

- Diabetic ketoacidosis:
  - insulin deficiency:
    - lipolysis is enhanced and there is an increase in the amount of fatty acids delivered to the liver.
      - The consequence is increased gluconeogenesis (generation of glucose from certain non-carbohydrate carbon substrates) contributing to hyperglycemia.
      - increases production of ketone bodies (beta-oxidation generates one acetyl CoA for every removal of a two-carbon group from the fatty acid chain.) by the mitochondria of the liver at a rate that exceeds peripheral use.
  - Increased glucagon
    - contribute to activation of gluconeogenesis

Clinical manifestation

- Polyuria and dehydration result from the osmotic diuresis due to hyperglycemia and ↑glucose in urine.
- Although the serum K^+ concentration may appear normal or elevated because of volume contraction (there is not a shift of K^+ out of the cell into the blood caused by metabolic acidosis with ketone bodies), there is the total body deficiency of K^+ due to:
  - polyuria
  - ↑ urinary anion of hydroxybutyric acid: It is not reabsorbed and its neg. charge attracts K^+ to urine)
  - shift K^+ to renal tubular cells support K^+ excretion.
- Others: Kussmaul respirations with acetone odor on the breath, postural dizziness, central nervous system depression, ketonuria, anorexia, nausea, abdominal pain and thirst.
- Treatment: intravenous administration of
  - insulin to decrease glucose levels
  - fluids to replace lost fluid volume.
  - K^+, $Na^+$and phosphorus
- After the administration of insulin, the concentration of β-hydroxybutyrate promptly begins to decrease, and after a slight increase, acetoacetate concentration also begins to decrease.

Davenpost diagram



https://www.youtube.com/watch?v=snp5oq-hLpg&t=237s

1. Normal
2. Compens Resp Acidosis
3. Pure Resp Acidosis
4. Pure Met Acidosis
5. Compens Metab Acidosis
6. Compens Resp Alkalosis
7. Pure Resp Alkalosis
8. Pure Met Alkalosis
9. Compens Met Alkalosis